Chemical Equilibrium

1. Starting from $dG = VdP - SdT$, derive the Gibbs-Helmholtz equation : (30 pts)

$$\left[\frac{\partial\left(\frac{G}{T}\right)}{\partial T}\right]_P = -\frac{H}{T^2}$$

Hint:

$$\frac{\partial(u \cdot v)}{\partial T} = v\frac{\partial u}{\partial T} + u\frac{\partial v}{\partial T}$$

2. For the following reaction at 25 °C,

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g)$$

	SO_2 (g)	O_2 (g)	SO_3 (g)
$\Delta_f \bar{G}^o$(kJ/mol)	-300.1	0	370.4

(a) Calculate the equilibrium constant (K_p) at 25 °C. (30 pts)

(b) Using the Gibbs-Helmholz equation to find $\Delta_r \bar{G}$ at 60 °C and hence K_p at 60 °C. (40 pts)